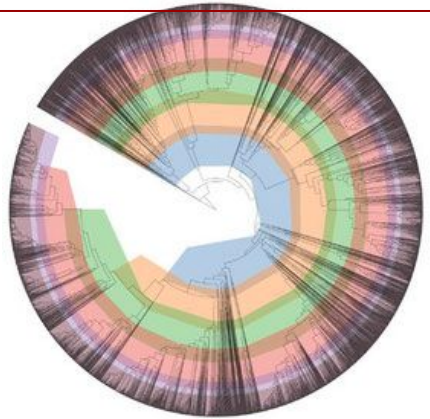
ENVIRONMENTAL METAGENOMICS

Physalia course, online, 12–16 October 2026

MAG functional annotation

Nikolay Oskolkov, Lund University, NBIS SciLifeLab
Luis Pedro Coelho, Queensland University of Technology



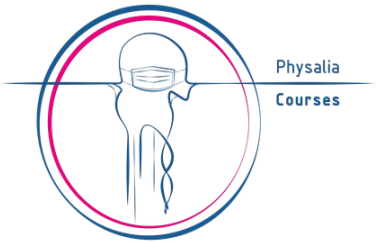
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NB: original course material courtesy:
Dr. Antti Karkman, University of Helsinki
Dr. Igor Pessi, Finnish Environment Institute (SYKE)

You got MAGs!
You know that they are good
Now what?

Functional annotation

- Genes
- *eggNOG*
- *RGI*
- *Other gene annotations databases*
- *…*

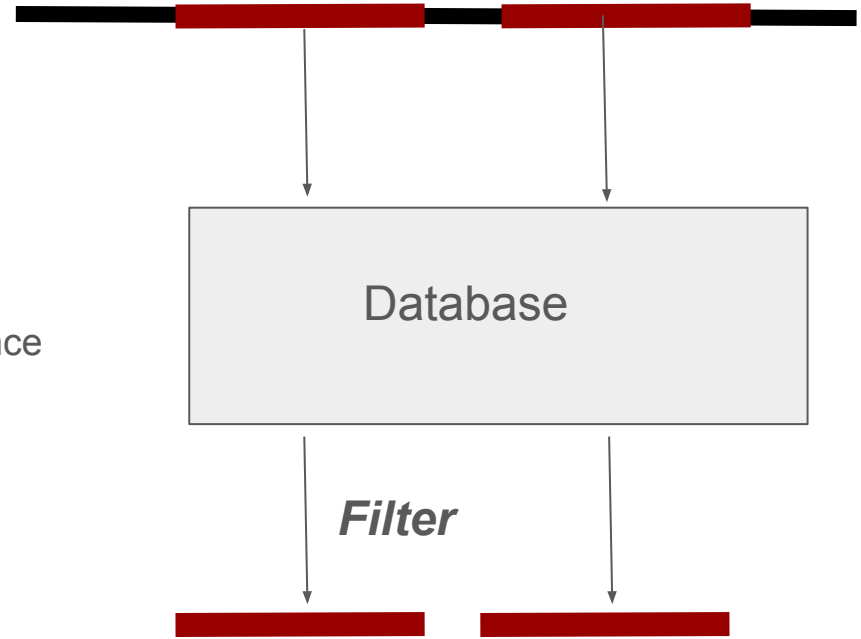




Congrats!
You got big FASTA files!

Functional annotation

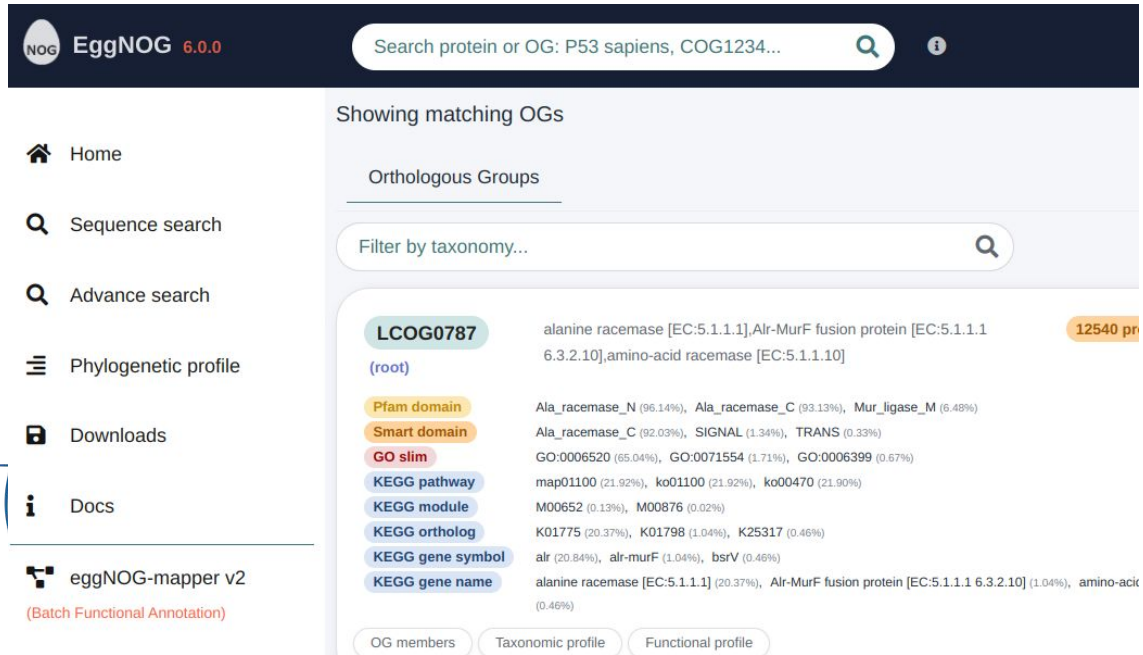
1. Predict genes
    a. Prodigal
    b. Pyrodigal
2. Predict function of genes
    a. Eggnog-mapper (generic)
    b. RGI (specialized to antibiotic resistance genes)
    c. …
3. Use informed human judgement!!

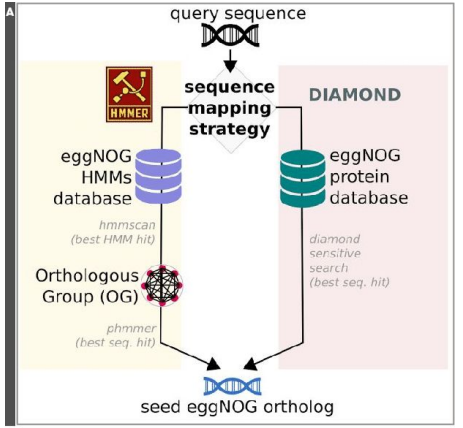


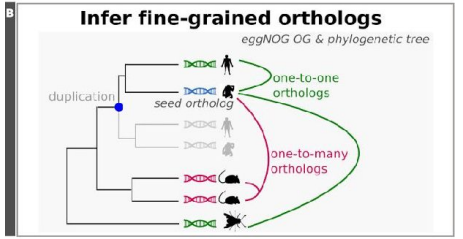
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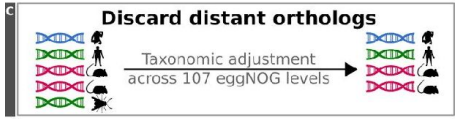
Gene annotation with eggnog-mapper

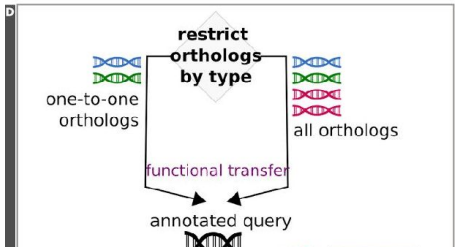
Basic concept: *eggNOG orthologous group*



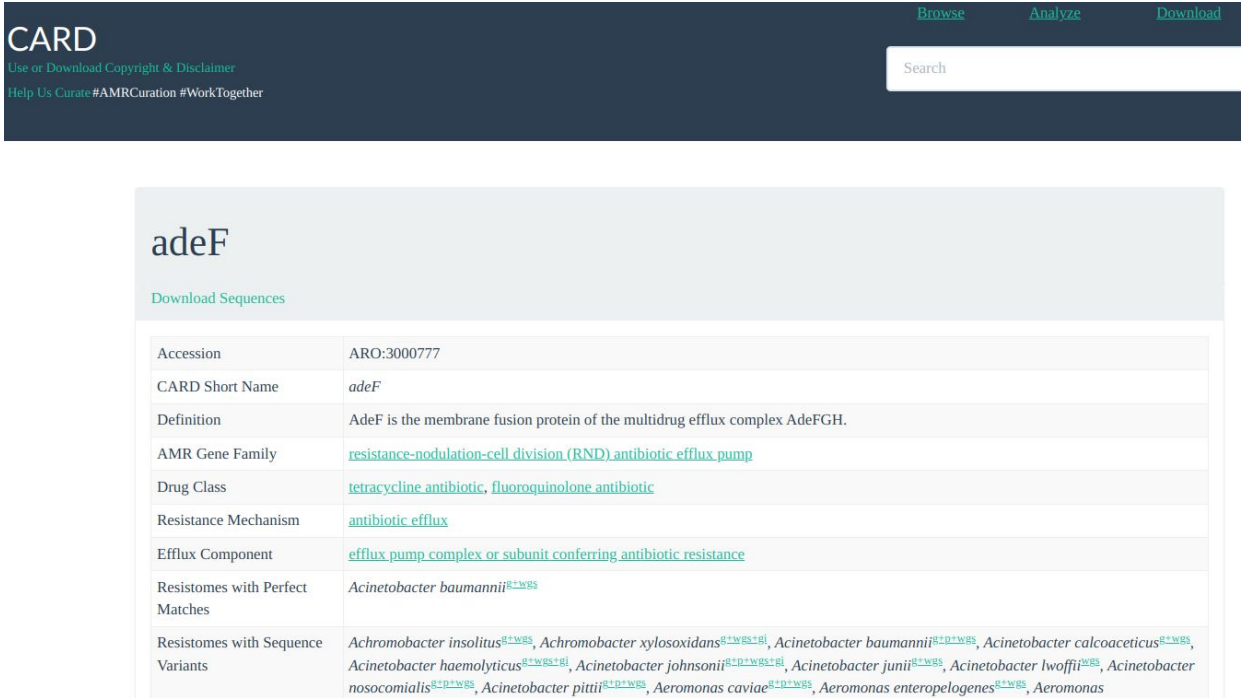








Gene annotation with RGI (CARD)



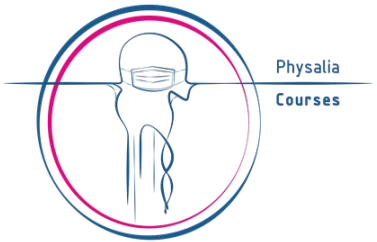
Similar principle: annotate to existing database of functions

Gene annotation with RGI (CARD): Annotation foibles

Strict: a match above threshold
Loose: a match below threshold
Nudged: a match below threshold, but high identity over a fragment

Beware the false positives!

- Efflux pumps!
- vanR (the regulator) without the regulatees
  - probably not ARG
- …



There are other specialized databases

- CAZy is very popular
- https://www.cazy.org/

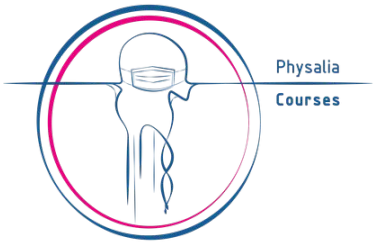


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What about small genes?

Traditionally ignored

We (and others) have started to look into it.



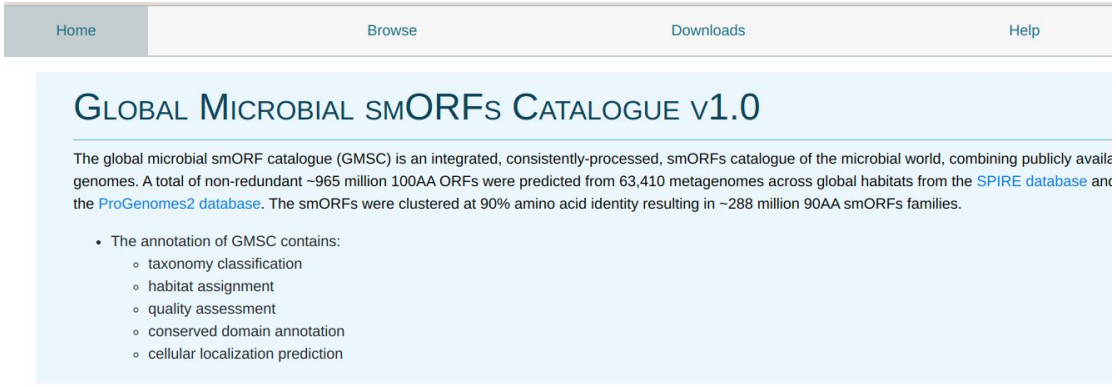
Article | Open access | Published: 31 August 2024

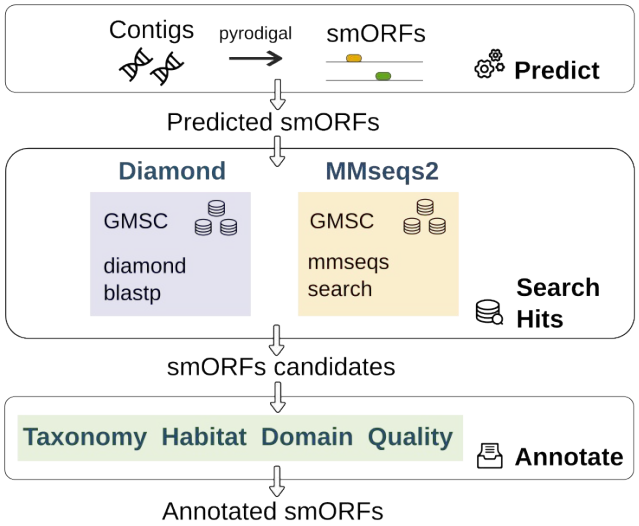
A catalog of small proteins from the global microbiome

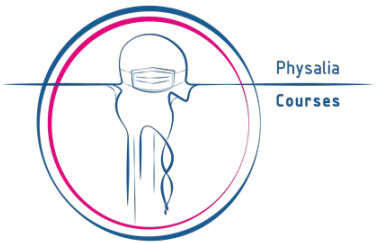
Yiqian Duan, Célio Dias Santos-Júnior, Thomas Sebastian Schmidt, Anthony Fullam, Breno L. S. de Almeida, Chengkai Zhu, Michael Kuhn, Xing-Ming Zhao ✉, Peer Bork & Luis Pedro Coelho ✉

Nature Communications **15**, Article number: 7563 (2024) | Cite this article

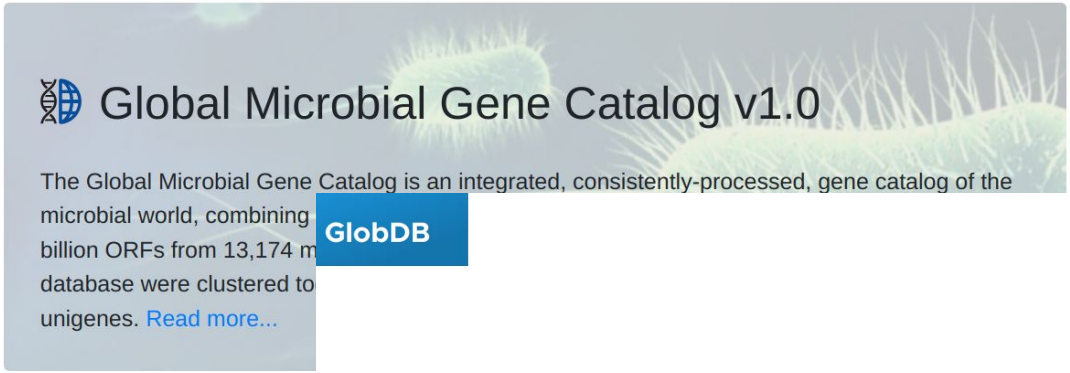
4348 Accesses | **58** Altmetric | Metrics

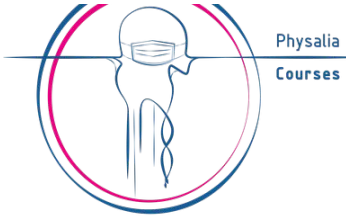

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GLOBAL MICROBIAL SMORFS CATALOGUE V1.0
The global microbial smORF catalogue (GMSC) is an integrated, consistently-processed, smORFs catalogue of the microbial world, combining publicly availa
genomes. A total of non-redundant ~965 million 100AA ORFs were predicted from 63,410 metagenomes across global habitats from the SPIRE database and
the ProGenomes2 database. The smORFs were clustered at 90% amino acid identity resulting in ~288 million 90AA smORFs families.
The annotation of GMSC contains:
taxonomy classification
habitat assignment
quality assessment
conserved domain annotation
cellular localization prediction
Search from identifier or find homologues by sequence (GMSC-mapper

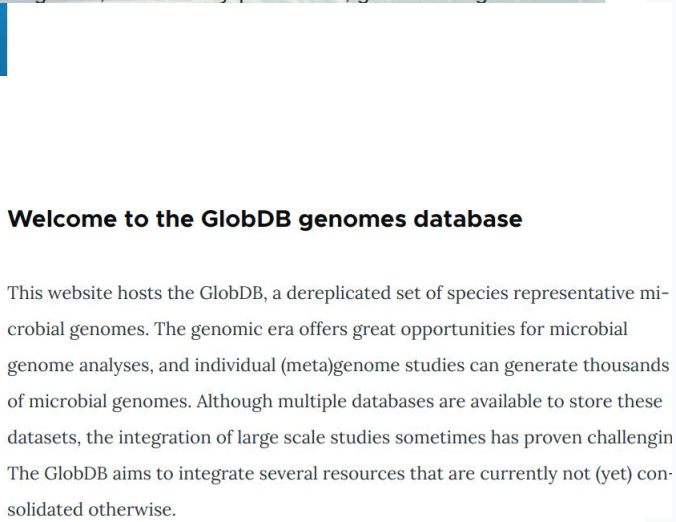

Contigs
pyrodigal
smORFs
Predict
Predicted smORFs
Diamond
MMseqs2
GMSC
diamond
blastp
GMSC
mmseqs
search
Search
Hits
smORFs candidates
Taxonomy Habitat Domain Quality
Annotate
Annotated smORFs

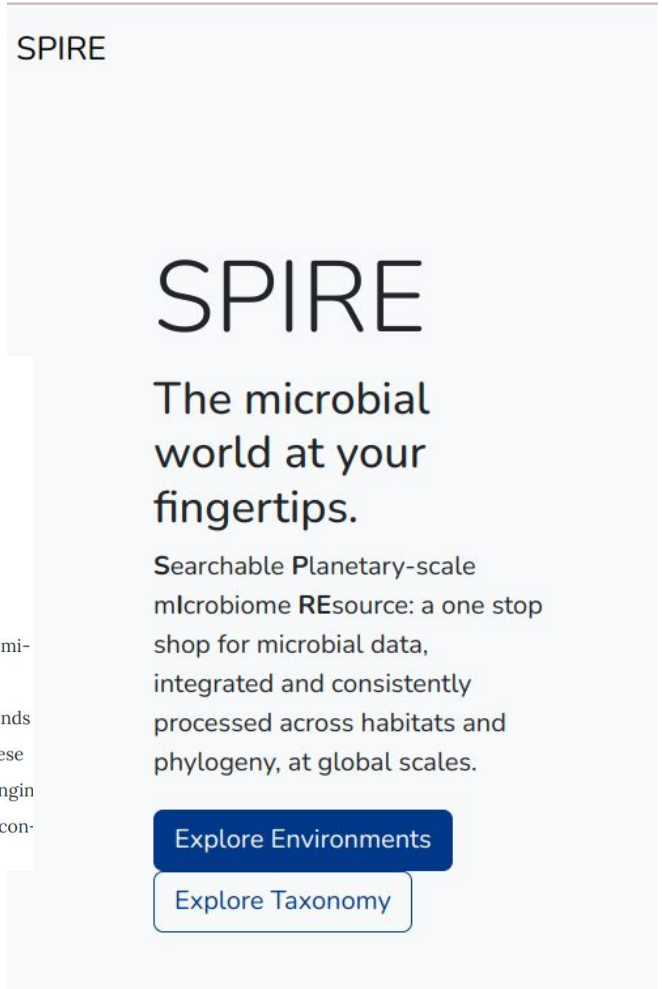

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Public databases of MAGs









What can you do with these databases?

1. Download their MAGs
2. Compare your MAGs to theirs (dereplication/pangenomes)
3. Check geographic distributions &c

